

Algorithm for file updates in Python

Project description

In this project, a secure allow list for a healthcare organization is maintained by a Python script. A file called `allow_list.txt`, which lists IP addresses permitted to access sensitive patient records, is updated automatically by the script. After reading the file and comparing its contents to a predetermined list of IP addresses that need to be deleted, the algorithm rewrites the file so that only the allowed IPs are left. This helps to keep patient data secure and guarantees that access rights are quickly terminated.

Open the file that contains the allow list

```
# Assign the file name to a variable
import_file = "allow_list.txt"

# Open the file
with open(import_file) as file:
    # Code to read the file will go here
```

Read the file contents

```
# Open the file in read mode
with open(import_file) as file:
    # Read the file contents
    ip_addresses = file.read()
```

Convert the string into a list

```
# Convert the string into a list
ip_addresses = ip_addresses.split()
```

Iterate through the remove list

```
# (Assuming remove_list is predefined)
# Iterate through the remove list
for element in remove_list:
    # Code to remove IPs will go here
```

Remove IP addresses that are on the remove list

```
# Iterate through the remove list
for element in remove_list:
    # Check if the element is in the ip_addresses list
    if element in ip_addresses:
        # Remove the element
        ip_addresses.remove(element)
```

Update the file with the revised list of IP addresses

```
# Convert the list back into a string with newlines
ip_addresses = "\n".join(ip_addresses)

# Write the updated string back to the file
with open(import_file, "w") as file:
    file.write(ip_addresses)
```

Summary

The update of an IP address allow list is effectively automated by this technique. The `allow_list.txt` file is first opened, and its contents are read into a string. After that, this string is transformed into a Python list so that it can be changed. An if statement inside the loop determines whether any of those IPs are in the main list, while a for loop iterates over a different `remove_list`. The `remove()` method removes the IP if a match is discovered. In order to safely replace the previous list, the "cleaned" list is finally transformed back into a string separated by newlines and written back to the original file.